



香港中文大學 (深圳)
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CSC 5003 Algorithm Art and Programming Practice

Lecture 2: Traverse on Trees and Graphs

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How do we store a tree/graph?

- ▶ Tree is a special form of graphs.
- ▶ There are directed/undirected/bidirected/unidirected graphs.
- ▶ Adjacent Matrix
- ▶ Adjacent List
- ▶ For binary tree, we simply store left,right children
 - How to store a tree in this way?



DFS & BFS

- ▶ Let's review on the definition of DFS/BFS first:
 - DFS: Depth-First, return whenever there is nowhere to go.
 - BFS: Breadth-First, level-by-level

- ▶ Implementation?

- ▶ Question: What can we know from a traverse on the graph?
 - Connectivity
 - A traverse order/tree
 - A recursion that can aggregate information from children.



Diameter Problem 1 (20 minutes)

- ▶ Given a binary tree, compute the longest path within.
- ▶ Be careful about the input!!!

Interview Hot 100



Diameter Problem 2 (10 minutes)

- ▶ Given an unrooted, unweighted tree, compute the longest path.
- ▶ Hint: Modify from the binary tree case!
- ▶ Any other method?



Diameter Problem 3 (10 minutes)

- ▶ Given a weighted binary tree, compute the maximum sum path.
- ▶ Hint: Modify from the unweighted case!
- ▶ What about the weighted tree version?



Summary

- ▶ A root can give us a direction to split the path and determine the parent-children relation
- ▶ We are doing dynamic programming here in fact.
- ▶ Try writing it into a recursion relation?



Path Sum iii (20 minutes)

- ▶ Given a binary tree and a target sum, find the number of down-paths such that the sum is equal to the target.



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- ▶ Tree is nice since it does not have cycles.
 - ▶ Can we solve some graph problems as if we are doing it on tree?
 - ▶ Yes! But please be very careful!!!!



Longest Cycle in a Graph (20 mins)

- ▶ Given a directed graph, find the largest cycle.
- ▶ Each vertex has at most one out-edge.
- ▶ Hint: what exactly does this graph look like?

Tencent



Shortest Cycle in a Graph

- ▶ Given an undirected graph, find the length of the shortest cycle.
- ▶ $N \leq 1000$
- ▶ What complexity is acceptable?



Find Eventual Safe States

- ▶ Given a directed graph, a vertex is "safe" if and only if you cannot walk into a cycle from it.
- ▶ Output all "safe" vertices.
- ▶ This is called "flood-fill".



Summary

- ▶ There are certain types of graph problems that we can decompose it into a tree problem + non-tree edges.
- ▶ Be careful about what to do once visiting a visited vertex (reach a back-edge)
- ▶ Draw some examples and observe!
- ▶ Advance: Learn the SCC algorithm called Tarjan!